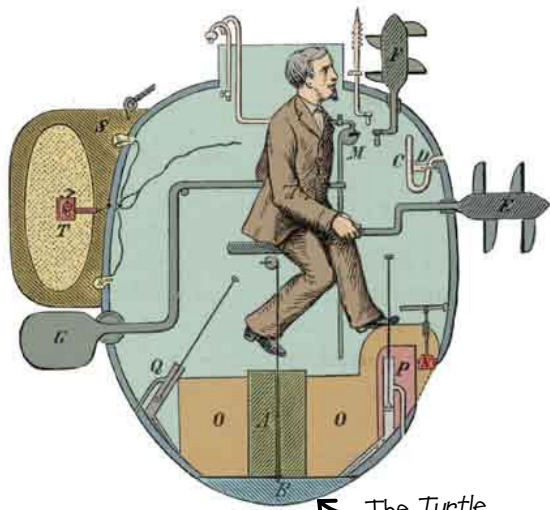




← The Turtle was driven by hand-operated propellers.

How it changed the world
Submarines changed the way wars were fought at sea by allowing crews to hide beneath the waves and launch surprise attacks on ships.

Early subs

The world's **FIRST SUBMARINE** dived beneath the surface of the Thames River in London in 1624. It was powered by 12 oars, and its crew **breathed oxygen** produced by heating potassium nitrate. The first sub to be used in warfare was the **Turtle** (left). It was used in 1776 during the American Revolution.

Submarine

DEEP-DIVING VESSELS that opened up an undersea world



USS Holland carried three torpedoes for underwater warfare.

USS Holland could dive 100 ft (30 m).

Holland's submarine

Submarines wouldn't have gone very far or very deep using manpower alone. In 1881, Irish-American engineer **J.P. Holland** demonstrated a submarine called the **Fenian Ram**, which used an engine on the surface and a battery when it dived. Holland had invented the modern submarine, and supplied the **US NAVY** with its first ever sub, **USS Holland**, in 1900.

Battle stations

Submarines launched **TORPEDOES** that sank hundreds of ships in World War I, and they've been used in **warfare** ever since. Today, most naval submarines use nuclear power and can stay **underwater for months** at a time, lurking in the depths of the ocean.

